Subspace-Constrained Byzantine Attacks

Weighted PCA Directions under Robust Aggregation

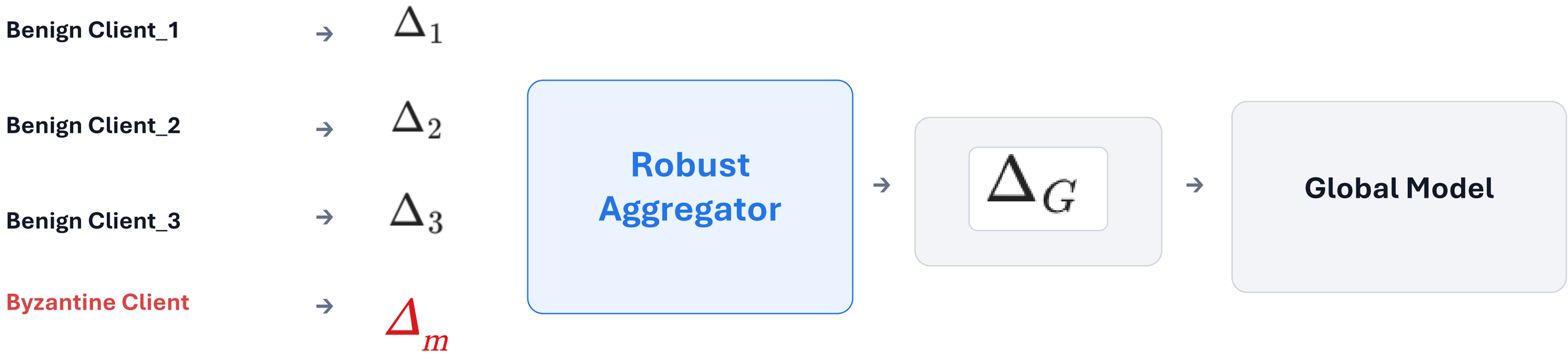
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Exploring structured perturbation directions beyond classical Byzantine attacks

Federated Learning with Byzantine Clients



FEDERATED LEARNING

Clients train locally and send updates Δ_i.

BYZANTINE THREAT

Malicious clients may submit arbitrary updates.

DEFENSE

Robust aggregators reduce malicious influence.

How should a Byzantine client construct a harmful malicious update?

Existing Byzantine Attack Strategies

ALIE

$$\Delta_m = \mu_b \pm z\sigma_b$$

Exploit benign coordinate-wise statistics.

Baruch et al., 2019

MIN-MAX

$$\Delta_m = \mu_b + \gamma d$$

Maximize γ under a maximum-distance constraint.

Shejwalkar & Houmansadr, 2021

MIN-SUM

$$\Delta_m = \mu_b + \gamma d$$

Maximize γ under a sum-distance constraint.

Shejwalkar & Houmansadr, 2021

$$\Delta_m = \mu_b + \underbrace{\gamma}_{\text{magnitude}} \underbrace{d}_{\text{direction}}$$

Malicious update construction can be viewed through magnitude and direction.

What About the Direction?

EXISTING MIN-MAX / MIN-SUM

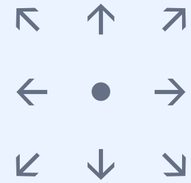
$$\Delta_m = \mu_b + \gamma d$$

γ: explicitly optimized

d: predefined candidates

$$d \in \left\{ -\operatorname{sign}(\mu_b), -\sigma_b, -\frac{\mu_b}{\|\mu_b\|_2} \right\}$$

? Could there be more harmful directions outside these predefined choices?



Can we systematically search for more harmful perturbation directions?

Constraining Direction Search with PCA

FULL SPACE

$$d \in \mathbb{R}^D$$

Neural-network parameter space is extremely high-dimensional.

Full search is intractable.

BENIGN GEOMETRY

Benign updates

$$X_i = \Delta_i - \mu_b$$

PCA

$$v_1, v_2, \ldots, v_r, \qquad r \leq N_b - 1$$

PCA SUBSPACE

$$\mathcal{S}_r = \mathrm{span}\{v_1, \ldots, v_r\}.$$

Data-driven, low-dimensional basis derived from benign variation.

PCA is not assumed to identify the most harmful direction.

It provides a structured space in which to search.

Weighted PCA Direction Search

$$d(K, \mathbf{s}) = \frac{\sum_{i=1}^{K} s_i w_i v_i}{\left\| \sum_{i=1}^{K} s_i w_i v_i \right\|_2}$$

$$s_i \in \{-1, +1\}, \qquad |\mathcal{D}_K| = 2^K$$

$$K = 2 \Rightarrow 4 \text{ directions}, \qquad K = 3 \Rightarrow 8 \text{ directions}$$

COMPONENTS

v_i	principal component	w_i	variance / std weight
s_i	component orientation	K	number of PCs

K = 3 SIGN SEARCH

(+,+,+) (+,+,−) (+,−,+) (+,−,−)

(−,+,+) (−,+,−) (−,−,+) (−,−,−)

PCA basis + weighted combination + component-wise sign search

Selecting the Most Disruptive Direction



$$\Delta_m(\mathbf{s}) = \mu_b + \gamma_{\mathbf{s}} d(K, \mathbf{s}) \qquad J(\mathbf{s}) = \|A(\Delta_b, \Delta_m(\mathbf{s})) - \mu_b\|_2$$

$$\mathbf{s}^* = \arg \max_{\mathbf{s} \in \{-1,+1\}^K} J(\mathbf{s}) \qquad \Delta_m^* = \mu_b + \gamma^* d(K, \mathbf{s}^*)$$

UPDATE-SPACE SELECTION

No test accuracy is used for direction selection.

AGGREGATOR-AWARE

The selected direction may differ across robust aggregation rules.

Experimental Setup

DATA & MODEL

- Dataset: Cifar 10
- Model: small CNN
- Non-IID split: dirichlet distribution with alpha set to 0.5

THREAT MODEL

- Clients: 10
- Byzantine: 3
- Rounds: 50

ATTACKS

- **Weighted PCA K=2**
- **Weighted PCA K=3**
- **ALIE**
- **Min-Max**
- **Min-Sum**

AGGREGATORS

- FedAvg
- Krum
- Multi-Krum
- Coordinate-wise median
- Timmed mean

Report variability across seeds

Last Five Round Average Accuracy without Nearest Neighbor Mixing

	No Attack	ALIE	MinMax	MinSum	SF	PCA k=2	PCA k=3
FedAvg	64.31 ± 1.26	40.33 ± 10.86	21.33 ± 19.63	36.82 ± 23.28	49.90 ± 0.94	49.17 ± 3.20	49.55 ± 3.81
Median	55.94 ± 3.11	21.29 ± 7.64	39.02 ± 2.78	41.52 ± 3.41	23.56 ± 1.25	33.82 ± 6.65	34.50 ± 2.86
TrimmedMean	64.34 ± 1.29	15.37 ± 4.94	38.72 ± 3.13	40.37 ± 3.99	21.38 ± 3.97	32.29 ± 2.19	27.44 ± 7.94
Krum	25.26 ± 17.69	10.00 ± 0.00	10.00 ± 0.00	9.99 ± 0.01	19.06 ± 15.69	10.00 ± 0.00	10.00 ± 0.00
MultiKrum	61.01 ± 1.60	13.79 ± 6.57	18.29 ± 14.37	10.00 ± 0.00	10.00 ± 0.00	10.00 ± 0.00	10.00 ± 0.00

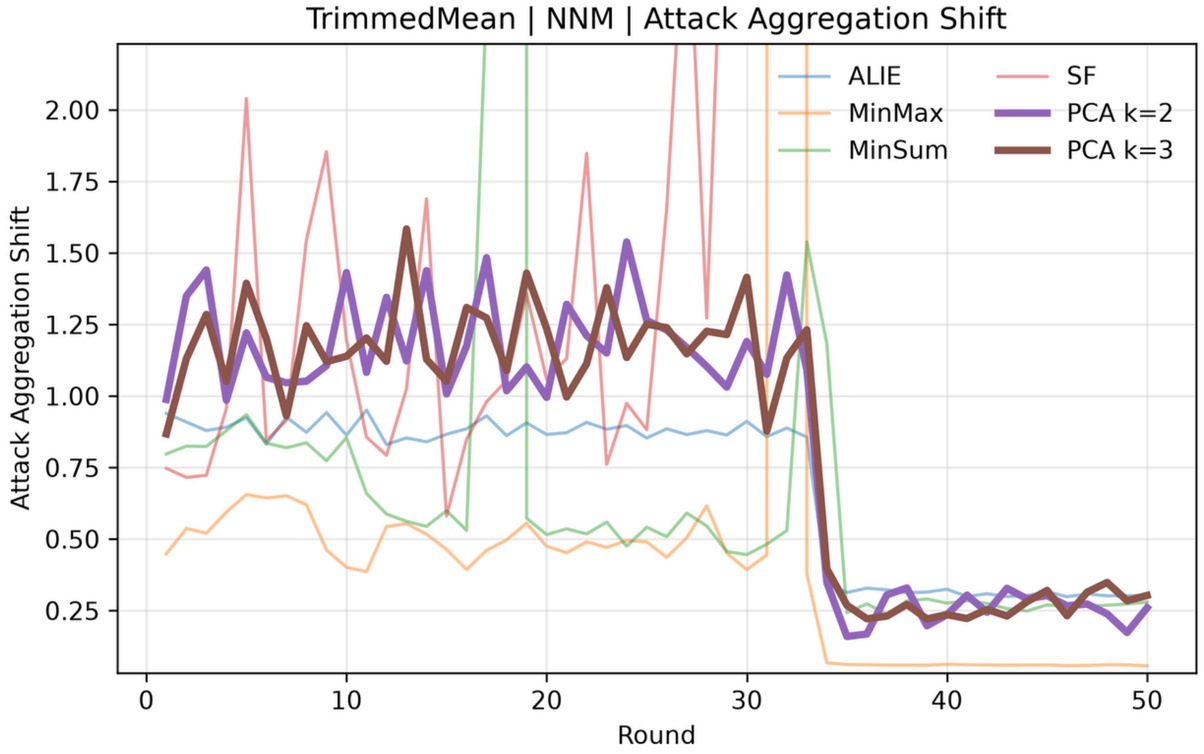
The most effective attacks were highlighted in yellow.

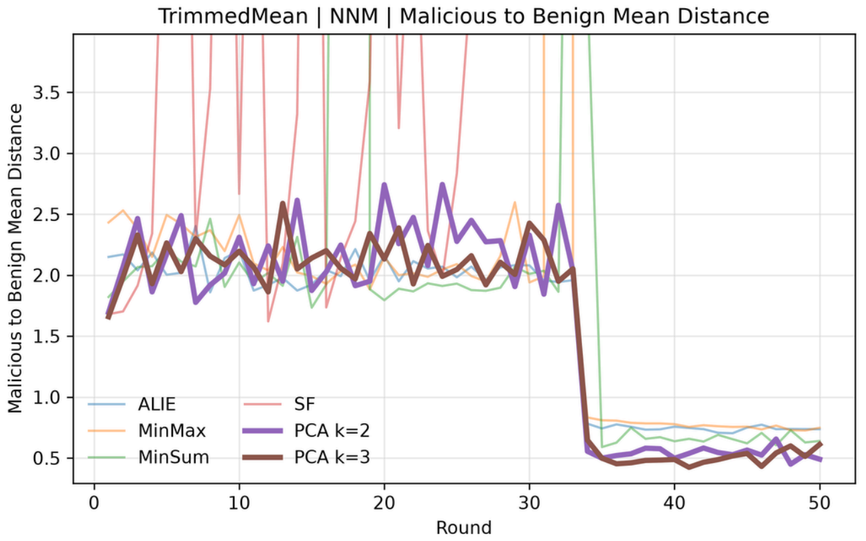
Last Five Round Average Accuracy with Nearest Neighbor Mixing

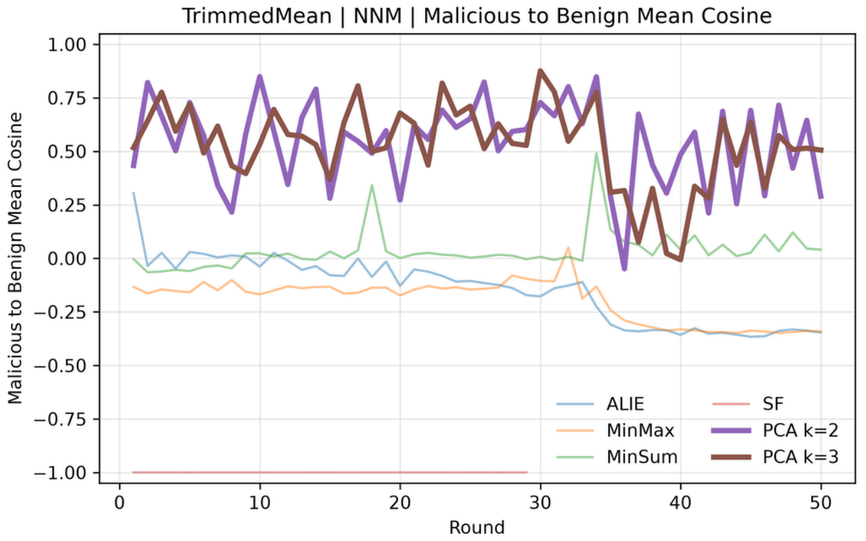
	No Attack	ALIE	MinMax	MinSum	SF	PCA k=2	PCA k=3
FedAvg	64.28 ± 1.22	48.94 ± 6.15	36.39 ± 22.99	34.19 ± 21.09	40.14 ± 7.15	47.94 ± 2.94	46.36 ± 5.10
Median	64.32 ± 1.32	37.29 ± 12.17	39.91 ± 25.91	34.78 ± 21.48	10.00 ± 0.00	29.14 ± 13.82	31.04 ± 2.27
TrimmedMean	64.33 ± 1.30	38.50 ± 11.30	40.54 ± 26.46	24.68 ± 16.44	10.00 ± 0.00	22.58 ± 9.04	21.92 ± 6.95
Krum	64.31 ± 1.27	35.56 ± 11.29	39.80 ± 25.82	23.26 ± 22.98	11.58 ± 2.74	18.44 ± 6.74	16.96 ± 6.12
MultiKrum	64.29 ± 1.25	41.21 ± 11.42	39.89 ± 25.89	20.78 ± 18.67	10.00 ± 0.00	27.16 ± 6.43	18.34 ± 7.28

Sign Flip is omitted from the main comparison because it consistently causes numerical failure (logit overflow) rather than stable degraded convergence.

Trimmed Mean with NNM







01 Direction selection matters

Optimizing perturbation magnitude alone is not sufficient; the choice of perturbation direction can substantially change attack effectiveness.

02 PCA provides a useful structured direction space

Weighted PCA reduces the high-dimensional direction search to a small, structured subspace, where component-wise sign search can identify more disruptive directions.

03 Defense preprocessing changes which directions are effective

PCA directions are not universally stronger, but under NNM they outperform the strongest stable classical baseline on 4 of 5 tested aggregation rules.

Next: move from discrete PCA direction search to adaptive direction optimization.

Questions

References

- Baruch, G., Baruch, M., and Goldberg, Y. (2019). A Little Is Enough: Circumventing Defenses For Distributed Learning. Advances in Neural Information Processing Systems (NeurIPS).
- Shejwalkar, V. and Houmansadr, A. (2021). Manipulating the Byzantine: Optimizing Model Poisoning Attacks and Defenses for Federated Learning. Network and Distributed System Security Symposium (NDSS).

Trimmed Mean without NNM

